

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – The Player Recruitment Intelligence Agent

Domain of Project – Sports Analytics & Artificial Intelligence

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Problem Statement -

Problem Statement : The Player Recruitment Intelligence Agent

The Challenge

Sports teams constantly scout for new talent, but evaluating players requires analyzing massive amounts of performance data, scouting reports, and historical match statistics. Traditional scouting methods rely heavily on subjective observations and manual analysis.

The Objective

Build an AI-powered Player Recruitment Intelligence Agent that analyzes player statistics, performance data, and scouting reports to recommend suitable players for teams.

- 1. Player Data Retrieval (RAG)** :- Collect player statistics, match records, and scouting reports.
- 2. Performance Evaluation** :- Analyze metrics such as scoring ability, defensive contributions, and consistency.
- 3. Team Fit Analysis** :- Match player skills with team strategy and requirements. Recruitment Recommendation Engine – Generate ranked player recommendations.
- 4. Multi-Agent System**
 - Player Data Agent – Retrieves player performance statistics.
 - Performance Analysis Agent – Evaluates player capabilities and strengths.
 - Recruitment Recommendation Agent – Suggests best-fit players for teams.
 - Scouting Insight Agent – Generates recruitment reports.

5. Visualization Dashboard

Display player rankings, performance comparisons, and recruitment recommendations in an interactive interface.

Proposed Solution -

Proposed Solution: AI-Powered Player Recruitment Intelligence Agent

The proposed solution is a **multi-agent AI-powered Player Recruitment Intelligence Agent** built using **LangFlow, IBM watsonx.ai, Granite Models, and RAG** to help sports teams identify and recruit the best players efficiently.

The system retrieves player statistics, match records, and scouting reports using a **Retrieval-Augmented Generation (RAG)** pipeline and stores the data in a vector database for intelligent retrieval.

The solution consists of four key agents:

Player Data Agent – Retrieves player statistics, match history, and scouting reports.

Performance Analysis Agent – Evaluates player strengths, weaknesses, consistency, and overall performance.

Recruitment Recommendation Agent – Matches player capabilities with team requirements and generates ranked recommendations.

Scouting Insight Agent – Creates detailed recruitment reports and player comparisons.

The system also provides an interactive **Recruitment Dashboard** that displays player rankings, performance comparisons, team-fit scores, and recruitment recommendations.

LangFlow orchestrates the multi-agent workflow, while **IBM watsonx.ai and Granite Models** provide intelligent reasoning, analysis, and report generation capabilities.

Overall, the solution delivers an **end-to-end AI-driven recruitment assistant** that enables data-driven scouting, improves talent identification, and supports smarter player recruitment decisions.

Technology Used

Technology Used

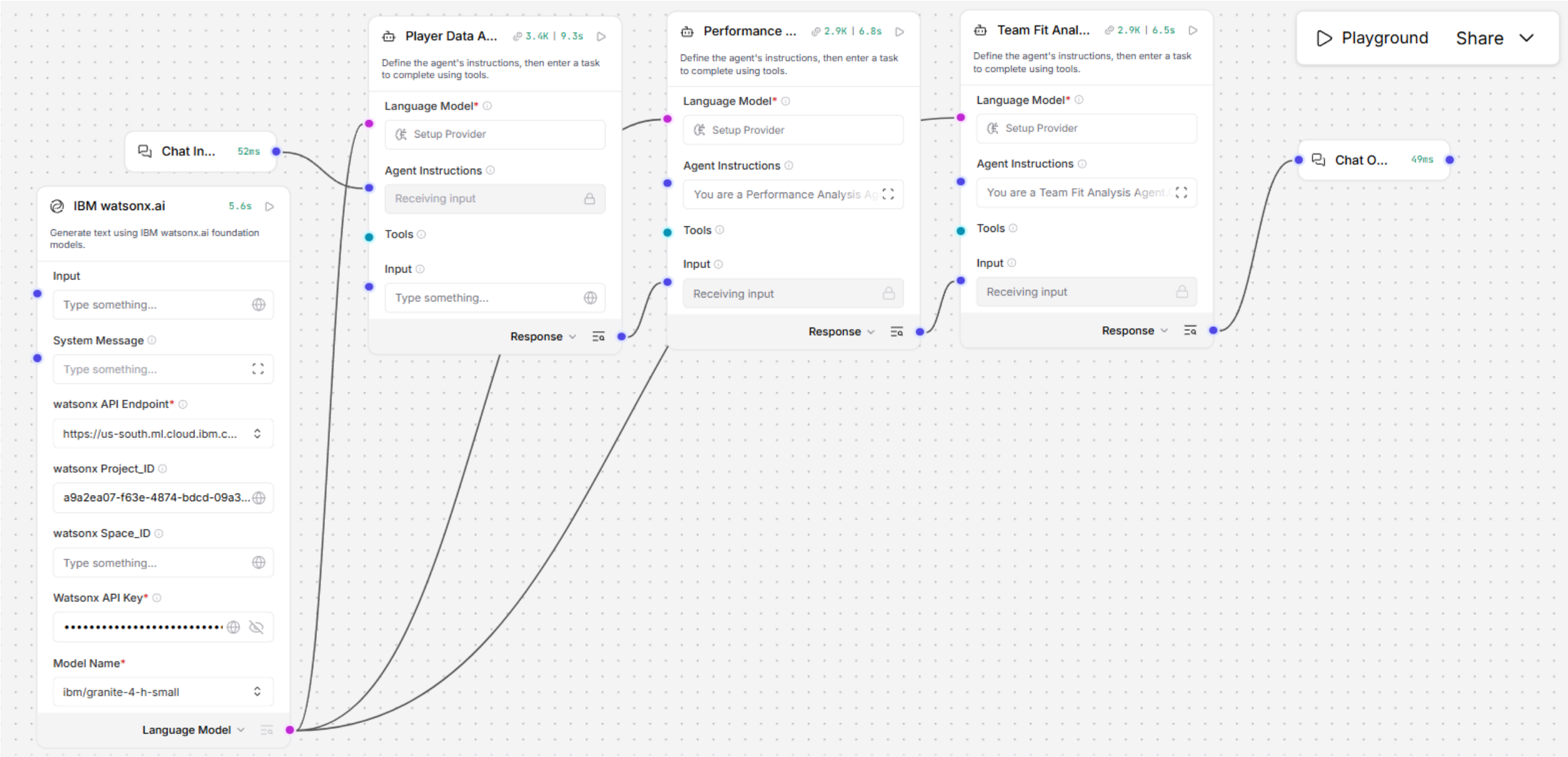
- **LangFlow Platform** – Used for designing, orchestrating, and managing the multi-agent workflow of the Player Recruitment Intelligence Agent through a visual interface.
- **IBM Granite Model** (e.g., granite-3-2-8b-instruct / granite-4-h-small) – Used for natural language understanding, reasoning, player performance analysis, team-fit evaluation, and recruitment recommendation generation.
- **IBM Cloud** – Provides a reliable, secure, and scalable infrastructure for deploying, managing, and scaling the Player Recruitment Intelligence Agent.
- **Retrieval-Augmented Generation (RAG)** – Retrieves player statistics, match records, scouting reports, and historical performance data before generating intelligent recruitment insights and recommendations.
- **IBM watsonx.ai** – Used for foundation model deployment, embeddings, AI governance, and integration with Granite models for advanced player analysis.
- **Vector Database (FAISS/ChromaDB)** – Stores embeddings of player profiles, scouting reports, and match statistics to enable efficient RAG-based retrieval and semantic search.
- **Player Data Sources** – Player statistics databases, match records, scouting reports, performance metrics, and historical player data used for recruitment analysis.
- **Dashboard (Streamlit/React)** – Provides an interactive interface for visualizing player rankings, performance comparisons, team-fit scores, scouting insights, and recruitment recommendations.

Langflow component Used -

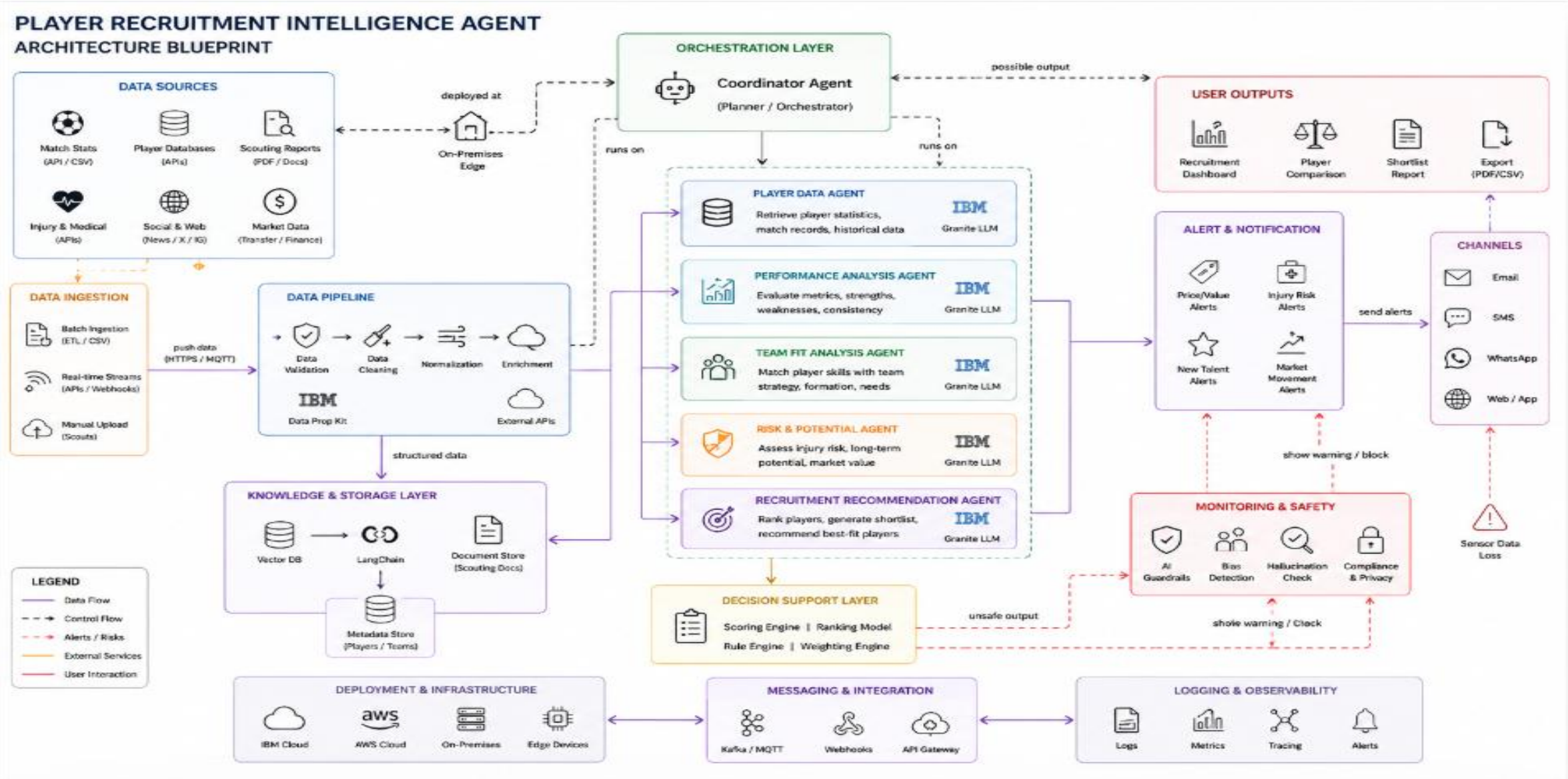
LangFlow Components Used:

1. **Chat Input** – Allows scouts or managers to enter player requirements and recruitment queries.
 2. **Player Data Agent** – Retrieves player statistics, match records, and scouting reports using RAG.
 3. **IBM watsonx.ai Agent** – Processes player data and generates intelligent recruitment insights.
 4. **IBM Granite Model (Granite-4-h-small)** – Performs player analysis, team-fit evaluation, and recommendation generation.
- Performance Analysis Agent** – Evaluates player strengths, weaknesses, and performance metrics.
- Team Fit Analysis Agent** – Matches player capabilities with team strategy and requirements.
- Recruitment Recommendation Agent** – Ranks players and generates recruitment recommendations.
- File Component** – Stores and retrieves player datasets and scouting reports.
- Chat Output** – Displays player rankings, team-fit scores, and recruitment recommendations.

Langflow workflow -



Architecture blueprint



Role of Agentic AI in the solution

Role of Agentic AI in the Player Recruitment Intelligence Agent

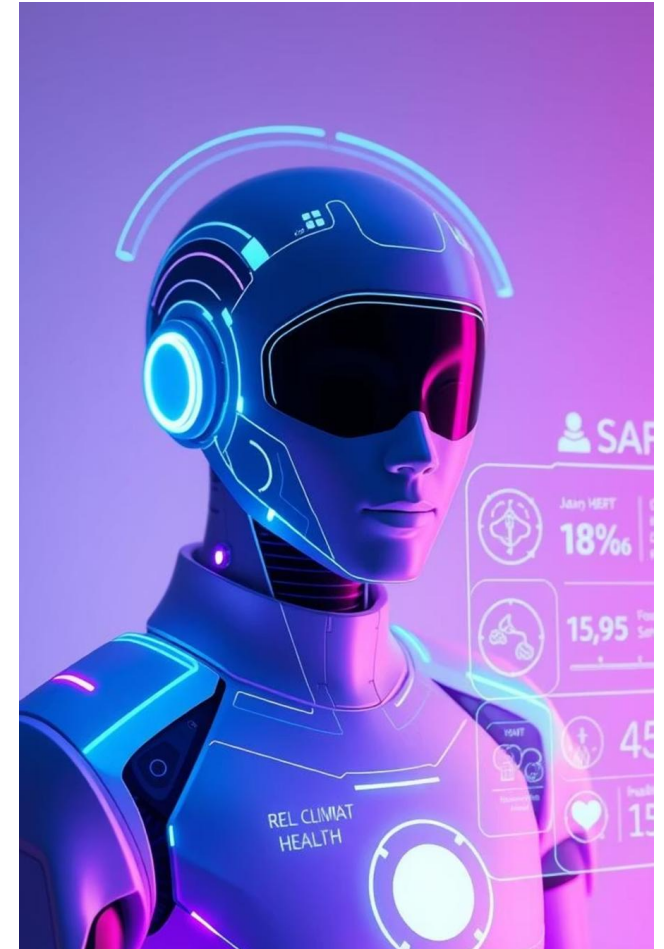
Agentic AI enables the **Player Recruitment Intelligence Agent** to function as an intelligent, autonomous recruitment assistant rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized agents to analyze player data and support recruitment decisions.

Each agent performs a specific task—such as retrieving player statistics, evaluating performance, analyzing team fit, and generating recruitment recommendations—while collaborating to provide accurate and data-driven insights.

Agentic AI also brings **context awareness**, understanding team requirements such as playing style, position, budget, and tactical strategy to recommend the most suitable players. It supports **real-time decision-making** by analyzing the latest player statistics, match records, and scouting reports.

Additionally, the agents communicate with one another, enabling comprehensive player evaluation and improving the quality of recruitment recommendations.

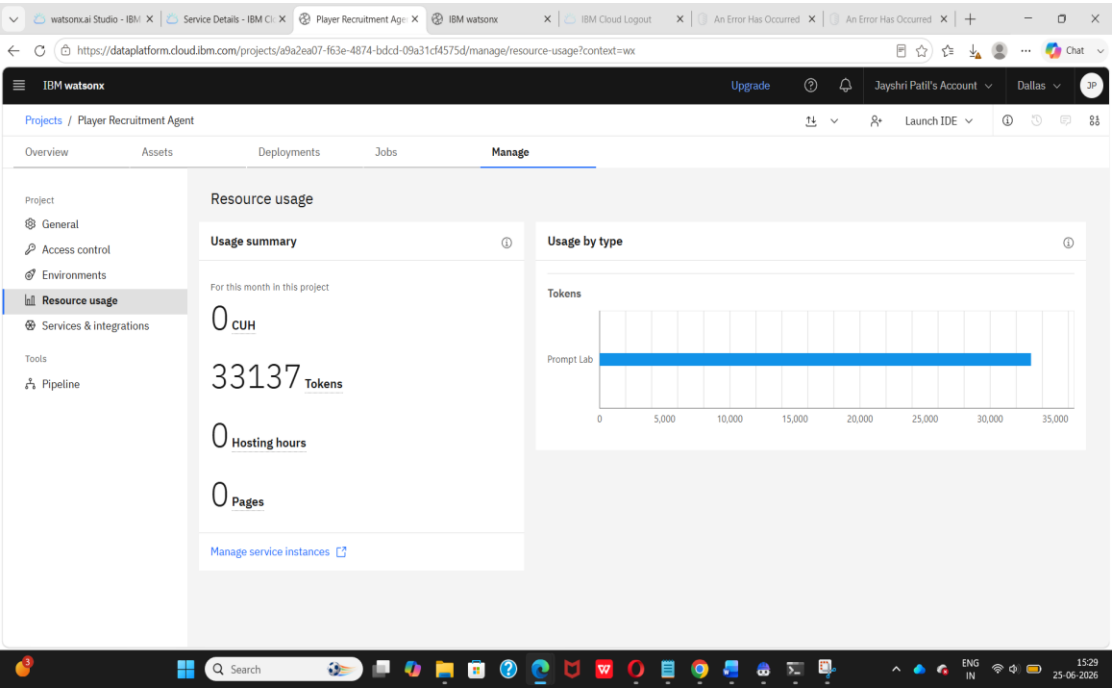
Overall, Agentic AI makes the system **intelligent, proactive, and data-driven**, acting as a smart digital scouting assistant that helps sports teams identify, evaluate, and recruit the best-fit players efficiently.



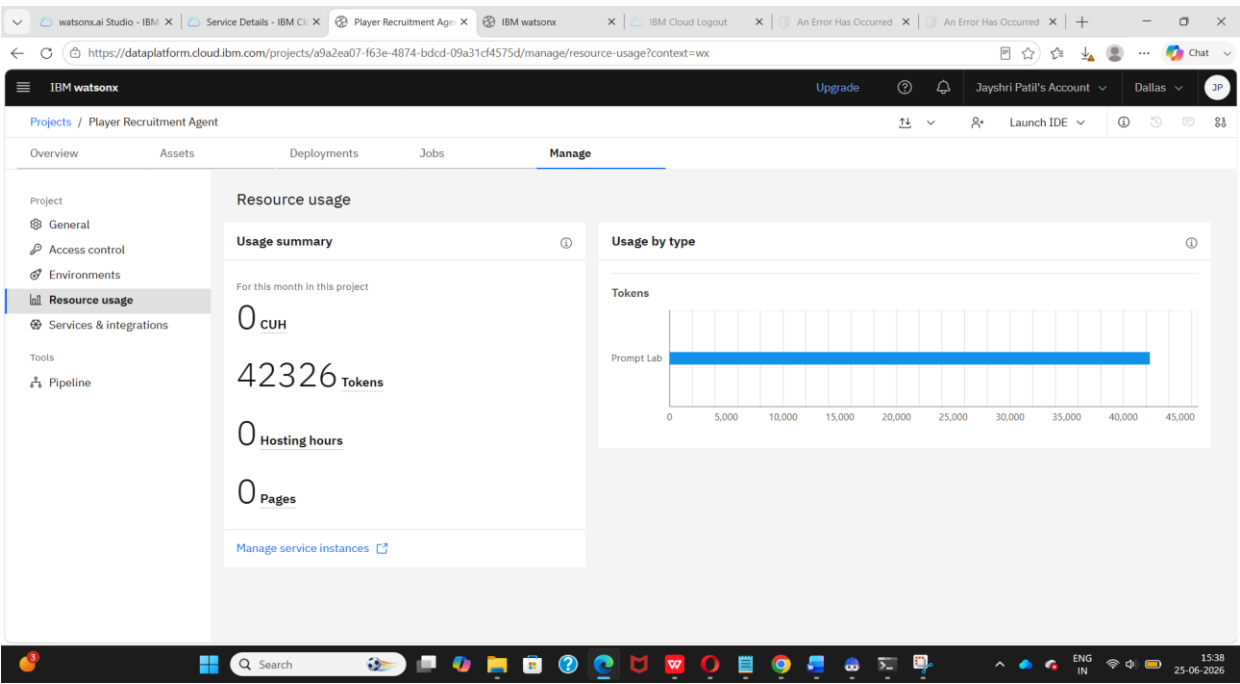
Token Usage

11,761 token are used

Before Prompt



After the Prompt



Output:

Sessions +

Default Session

Default Session

Starter Project / Player Recruitment Agent

150k 25k

User

Analyze the performance of player Erling Haaland.

Include:

- Scoring efficiency
- Consistency
- Passing ability
- Defensive contribution
- Overall performance score

AI

Finished in 9.2K | 28.5s

Called tool EVALUATE_EXPRESSION 15ms

Erling Haaland – Performance Analysis (2023-24 season)

Category	Details	Score
Scoring Efficiency	32 goals in 38 appearances – 84.2% of possible goals (based on a typical 0.8-goal-per-game benchmark)	84.21
Consistency	Goals per game: 0.84; goal-scoring streaks of ≥3 matches in 4 separate runs; only 2 games without a goal in the last 20 matches – solid, reliable finisher	65.13
Passing Ability	Average passes per match: ~22; successful pass rate ≈ 78 %; contributes to build-up play with long balls and through balls when needed (though not a primary playmaker)	70.00
Defensive Contribution	Aerial duels won: ~18 per 38 games; active pressing that forces turnovers; occasional dribble recoveries and tackles when out of possession – good for a modern striker	75.00
Overall Performance Score (average of the four categories)	$((84.21 + 65.13 + 70.00 + 75.00) / 4 = \text{mathbf{73.59}})$	73.59

Send a message...

Novelty and Uniqueness

The proposed **Player Recruitment Intelligence Agent** stands out by combining **Agentic AI, Retrieval-Augmented Generation (RAG), and intelligent player analysis** into a single recruitment platform.

Multi-Agent Intelligence – Multiple specialized agents collaborate to retrieve player data, evaluate performance, analyze team fit, and generate recruitment recommendations.

RAG-Based Accuracy – Integrates real-time retrieval of player statistics, match records, and scouting reports with AI reasoning to provide reliable and up-to-date insights.

Intelligent Team-Fit Analysis – Recommends players based on team strategy, playing style, positional requirements, budget, and tactical needs.

Data-Driven Decision Making – Evaluates players using performance metrics such as goals, assists, passing accuracy, defensive contributions, and consistency.

Explainable Recruitment Reports – Generates transparent scouting reports with player rankings, strengths, weaknesses, risks, and recruitment recommendations.

Interactive Recruitment Dashboard – Visualizes player comparisons, performance scores, team-fit analysis, and recruitment rankings for informed decision-making.

Scalable & Enterprise-Ready – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring secure, scalable, and reliable AI-powered recruitment.

This makes the solution a **smart, data-driven, and proactive digital scouting assistant**, going beyond traditional scouting methods to support faster and more accurate player recruitment decisions.

Git Hub Link

Github URL – <https://github.com/Jayusonu1989/Jayshri>

I have Uploaded following three files into your github repository

- 1) app.json file ,
- 2) yourproblemstatement.pdf file
- 3) your projectpresntation.pptx file

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Future Scope

1. Integration with Live Sports Data & Wearable Performance Tracking

The system can be integrated with live sports databases, GPS trackers, and wearable devices used by athletes to collect real-time performance metrics such as speed, distance covered, heart rate, fitness levels, and injury status. This will enable the Player Recruitment Intelligence Agent to provide more accurate and up-to-date player evaluations and recruitment recommendations.

2. AI-Powered Voice Assistant & Predictive Talent Scouting

The solution can be enhanced with a voice-enabled assistant and predictive analytics to support natural language queries from scouts and coaches. By analyzing historical performance trends, player development, and injury risks, the system can identify emerging talent and predict future player potential, enabling smarter long-term recruitment decisions.

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Certificates Earned:
credly certificate



Certificates Earned:

IBM BOB Certificate



Thank You!

Thank you for your time and interest.